

Are Proof Structures Machine-Learnable?

Follow-up work on Generating Theorems by Generating Proof Structures

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From proof compression to generative lemma synthesis

Previous direction

Lemma synthesis via proof enumeration and compression

Current direction

Lemma synthesis via a generative language model

Can we learn to generate useful lemmas
instead of systematically enumerating them?

Example proof term

$D(D(D(\textcolor{red}{2}, D(\textcolor{blue}{1}, \textcolor{blue}{1})), D(\textcolor{red}{2}, D(\textcolor{blue}{1}, \textcolor{blue}{1}))), D(\textcolor{blue}{1}, \textcolor{blue}{1}))$

The proof term as DAG grammar

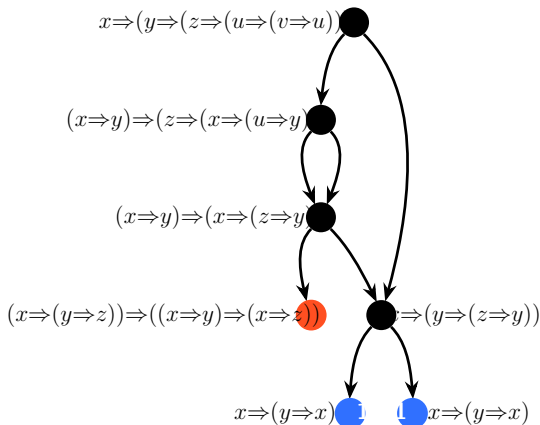
$4 \rightarrow D(\textcolor{blue}{1}, \textcolor{blue}{1})$

$5 \rightarrow D(\textcolor{red}{2}, 4),$

$Start \rightarrow D(D(5, 5), 4)$

Compact Prefix Notation

$D(s = ddX5X5X4$
 $|X4 = dax1ax1$
 $|X5 = dax2X4$
 $)$



Why generate lemmas? – Two possible use cases

Theory exploration

- ▶ Generate mathematically interesting true statements
- ▶ Recover human-style lemmatization automatically
- ▶ Discover novel and useful theorems

Aiding proof search

- ▶ Generate lemmas useful for a particular target
- ▶ Similar to premise selection, but more flexible
- ▶ Not restricted to previously stated theorems
- ▶ Avoids searching through a large theorem database

Key distinction

For proof search, generation is conditioned on the target theorem.

Where does our current work fit?

Theory exploration

- ▶ No target theorem is given
- ▶ The model generates proofs from the theory

Proof search

- ▶ Generated lemmas are evaluated by their effect on ATP performance
- ▶ Usefulness is measured relative to proof problems

Our current setup is hybrid.

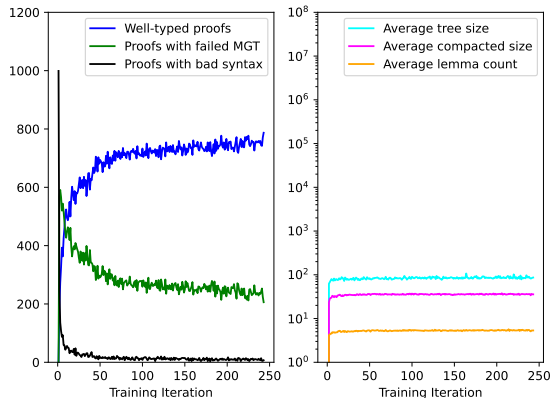
We would like feedback on which direction to develop further.

Project overview

1. Train a “small” generative language model from scratch (130 million parameters)
2. Pre-train with a large corpus of simple proofs
 - ▶ Use *SGCD* theorem prover to enumerate proof terms
 - ▶ 51 million proof terms, mostly uninteresting
 - ▶ Learn the proof syntax
 - ▶ Learn to generate well-typed proofs
3. Fine-tune on a small set of proofs from *set.mm* of *Metamath*
 - ▶ 1,147 proofs
 - ▶ Shift distribution towards proofs of interest
4. Evaluate the generated proofs and lemmas

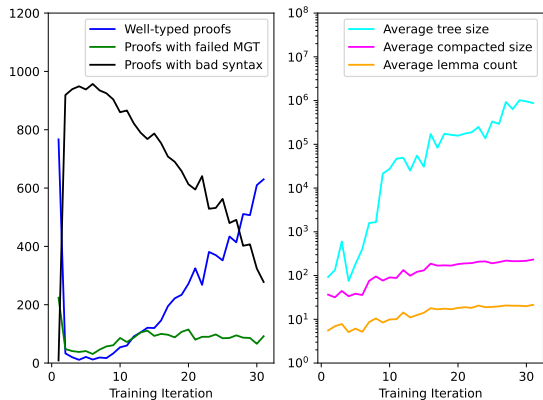
	#Proofs	Average			Median		
		Tsize	Csize	#Lem	Tsize	Csize	#Lem
Pre-training	5.1×10^7	133	36	6	69	36	5
Fine-tuning	1,147	2×10^{12}	297	30	11,833	279	23

What is learned during pre-training?



- ▶ Proof syntax is learned quickly.
- ▶ Well-typedness is harder to learn.
- ▶ By the end of pre-training, generation quality is already reasonable.
- ▶ Generated proofs are small

What does fine-tuning add?

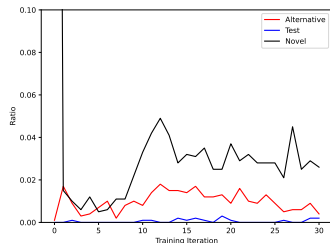


- ▶ The fine-tuning set contains only 1,147 proofs.
- ▶ Fine-tuning is needed to produce larger proofs.
- ▶ It also increases the structural complexity of generated proofs.

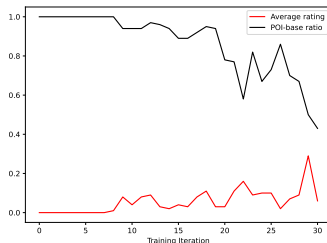
Interpretation

Pre-training teaches the model how to speak the proof language.
Fine-tuning moves it toward the proof distribution of interest.

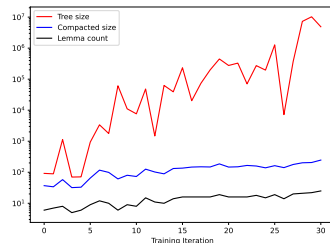
Evolution of generated proof types during fine-tuning



Well-typed proofs



Alternative and Test proofs



Novel proofs

- ▶ Training proofs: proofs seen during training
- ▶ Alternative proofs: new proofs of known statements
- ▶ Test proofs: proofs corresponding to held-out examples
- ▶ Novel proofs: proofs of statements not present in the corpus

Are the generated lemmas useful?

Preliminary evaluation

We do not yet know how to compare generated proofs properly with human proofs.

- ▶ Measure whether generated lemmas help an ATP prove theorems.
- ▶ 1,374 theorems *Metamath* POI theorems, 1,147 used for training, 60s
- ▶ Prefer lemmas with smaller most general theorem (MGT)
- ▶ Collect **Novel** theorems from the language model
- ▶ Augment axioms with lemmas

Prover	Base	Fine-tuning			Pre-training		
		TOP-175	TOP-100	TOP-10	PRE-175	PRE-100	PRE-10
<i>Vampire</i>	585 (48)	608 (42)	604 (46)	865 (66)	575 (38)	650 (48)	518 (37)
<i>E-auto</i>	692 (48)	415 (27)	442 (27)	446 (27)	293 (20)	293 (19)	273 (18)
<i>E-as</i>	655 (48)	386 (26)	417 (27)	497 (31)	302 (22)	445 (30)	392 (26)
<i>leanCoP</i>	69 (5)	327 (29)	233 (22)	168 (13)	188 (14)	151 (10)	65 (2)

Discussion and open directions

- ▶ How should we evaluate the quality of generated lemmas?
- ▶ Which properties make a lemma useful for proof search?
- ▶ Can we distinguish genuinely novel proofs from reformulations?
- ▶ Should generation be conditioned on a target theorem?
- ▶ What would make this a useful theory-exploration system?

We are interested in ideas for both directions: theory exploration and proof search.